



△ VAST SLAB

Parts of the Canadian Shield—a huge slab of tough and very ancient volcanic and metamorphic rocks—are exposed by erosion in areas such as riverbanks.



◁ SCULPTED PEAKS

One of the world's greatest mountain ranges, the Rockies run all the way along the western edge of North America. They have been heavily sculpted by glaciers.



THE SHAPING OF NORTH AMERICA

North America began life as part of the great, ancient landmass geologists call Laurentia, which journeyed across the world, colliding with and separating from other continents. The continent we know today came into being less than 200 million years ago.

Ages of rock

Underlying much of North America is a large shield of ancient rock that once formed the core of Laurentia. In the U.S., it is covered by a blanket of more recent sedimentary rock. But in Canada, where the shield is exposed on the surface, geologists have found some of the world's oldest rocks—survivors from the dawn of Earth's story, up to 4.4 billion years old.

About 750 million years ago, this Laurentian heartland formed part of the supercontinent of Rodinia, but after Rodinia broke up Laurentia drifted almost to the South Pole before drifting north again. For a while, Laurentia was alone, but eventually it was drawn into the regathering of continents and became part of a new supercontinent—Pangaea.

About 480 million years ago, the plate Laurentia sat on crunched into neighboring ocean plates, causing the Appalachian Mountains to start rising. It took 250 million years for the mountains to reach

their maximum height. At first, they were probably as tall and majestic as the Himalayas. In the southwest, rocks in Nevada and Utah were twisted by the same tectonic forces.

America's backbone

About 200 million years ago, Laurentia split from Pangaea to drift northwest, opening up the Atlantic Ocean and creating North America. The new continent's east coast settled down as it moved away, and the Appalachians were worn gradually lower and lower by millions of years of erosion and weathering. This is why the Appalachians are smaller than the Rocky Mountains even though they are older.

The west, on the other hand, became a titanic battleground between opposing tectonic plates, as the North American Plate plowed into and rode over the oceanic crust of the Pacific Plate, forcing it to subduct, or descend into the mantle. The Rocky

The Appalachians are over 480 million years old

KEY EVENTS

480 million years ago

The Appalachians start to rise. They continue rising for another 250 million years to Himalayan heights, but have since been lowered by erosion.

80-55 million years ago

The rising Rocky Mountains cause the surrounding land to uplift. As a result, the Western Interior Seaway, a large inland sea, dries out.

18 million years ago

The Greenland ice sheet forms around this time. However, it may be even older than this, and it is likely the ice sheet has receded and advanced many times since then.

11,000 years ago

The last of the ice sheets covering much of North America start to recede, relieving most of the continent of its covering of ice.

1.5-1 billion years ago Most of the land comprising the crust of central and eastern North America today forms.

200 million years ago

Tectonic activity splits Laurentia apart from Europe, allowing the land that will become North America to drift northwest.

26-22 million years ago The Earth's climate becomes cooler, favoring the spread of grasslands. The Great Plains first appear, covering a swathe of flat land that used to be the Western Interior Seaway.

15,000 years ago, During the last ice age, some glaciers start to melt, and water pools to form the Great Lakes.

10,000 years ago Ice melts and sea levels rise, causing the land bridge connecting Alaska and Russia to submerge.